WORKED SOLUTION

STEP III 1992, Question 1

Pure • Maclaurin Series

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s3-q1

The key idea

In part (i) the two identities are a machine for producing derivatives: once f'' is written in terms of f', every higher derivative follows by differentiating again, and at $x = 0$ everything is built from $f'(0) = \frac{1}{2}$. In part (ii) the trouble is the factor $\sinh x$, which vanishes at the origin; multiplying by x cancels that zero, and the expansion is then the product of two series we can find by hand. Symmetry (odd and even functions) tells us in advance which powers can appear.

Solution

Part (i)

Differentiating $f(x) = \ln(1 + e^x)$ by the chain rule,

$$f'(x) = \frac{e^x}{1 + e^x} > 0,$$

so its logarithm is defined, and

$$\ln[f'(x)] = \ln e^x - \ln(1 + e^x) = x - f(x),$$

which is the first result. Differentiating this identity, the left-hand side gives $f''(x)/f'(x)$, so

$$\frac{f''(x)}{f'(x)} = 1 - f'(x) \quad \Longrightarrow \quad f''(x) = f'(x) - [f'(x)]^2,$$

as required. It is convenient to write this as $f'' = f'(1 - f')$.

? **Ask yourself** How do we get f''' and f'''' without differentiating $\ln(1 + e^x)$ four times?

Differentiate the identity $f'' = f' - (f')^2$ itself, using the product rule on $(f')^2$:

$$f''' = f'' - 2f'f'' = f''(1 - 2f'),$$
$$f'''' = f'''(1 - 2f') - 2(f'')^2.$$

Now evaluate at $x = 0$, where $f(0) = \ln 2$ and $f'(0) = \frac{1}{1+1} = \frac{1}{2}$:

$$f''(0) = \tfrac{1}{2} - \tfrac{1}{4} = \tfrac{1}{4},$$
$$f'''(0) = \tfrac{1}{4}\left(1 - 2 \cdot \tfrac{1}{2}\right) = 0,$$
$$f''''(0) = 0 - 2\left(\tfrac{1}{4}\right)^2 = -\tfrac{1}{8}.$$

By Maclaurin's theorem, $f(x) = \sum f^{(n)}(0)\, x^n / n!$, so

$$f(x) = \ln 2 + \frac{x}{2} + \frac{x^2}{8} - \frac{x^4}{192} + \cdots,$$

the x^3 term being zero (and $-\frac{1}{8} \div 24 = -\frac{1}{192}$).

Part (ii)

Why g has no such expansion. A series $a_0 + a_1 x + a_2 x^2 + \cdots$ in non-negative powers takes the finite value a_0 at $x = 0$, and is continuous there. But $\sinh 0 = 0$ and $\cosh 0 = 1$, so the denominator $\sinh x \cosh 2x$ tends to 0 as $x \to 0$; indeed, since $\sinh x \approx x$ for small x, we have $g(x) \approx 1/x$, which is unbounded near 0. So $g(x)$ cannot equal such a series near the origin.

Why $xg(x)$ does. Write

$$x\, g(x) = \frac{x}{\sinh x} \cdot \frac{1}{\cosh 2x}.$$

Since $\sinh x = x\left(1 + \frac{x^2}{6} + \frac{x^4}{120} + \cdots\right)$, the factor x cancels and

$$\frac{x}{\sinh x} = \frac{1}{1 + \frac{x^2}{6} + \frac{x^4}{120} + \cdots},$$

which is 1 over a series whose constant term is 1; such a reciprocal can be expanded binomially in non-negative powers. Likewise $\cosh 2x \geq 1$ never vanishes and $1/\cosh 2x$ expands in the same way. The product of two such series is again one.

Why only even powers. sinh is odd and cosh is even, so $g(-x) = -g(x)$: g is odd. Hence $h(x) = x\, g(x)$ satisfies $h(-x) = (-x)(-g(x)) = h(x)$, so h is even. If $h(x) = \sum a_n x^n$, then $h(x) - h(-x) = 2\sum_{n \text{ odd}} a_n x^n$ is identically zero, and since a power series that vanishes identically has all coefficients zero, $a_n = 0$ for every odd n. (You can also see it directly: both factors above are series in x^2.)

The expansion. Using $(1+u)^{-1} = 1 - u + u^2 - \cdots$ with $u = \frac{x^2}{6} + \frac{x^4}{120}$, and keeping terms up to x^4,

$$\frac{x}{\sinh x} = 1 - \frac{x^2}{6} - \frac{x^4}{120} + \frac{x^4}{36} + \cdots = 1 - \frac{x^2}{6} + \frac{7x^4}{360} + \cdots,$$

since $\frac{1}{36} - \frac{1}{120} = \frac{10-3}{360}$. Next, $\cosh 2x = 1 + 2x^2 + \frac{2}{3}x^4 + \cdots$ (from $\frac{(2x)^2}{2}$ and $\frac{(2x)^4}{24}$), so with $v = 2x^2 + \frac{2}{3}x^4$,

$$\frac{1}{\cosh 2x} = 1 - v + v^2 - \cdots = 1 - 2x^2 - \tfrac{2}{3}x^4 + 4x^4 + \cdots = 1 - 2x^2 + \tfrac{10}{3}x^4 + \cdots.$$

Multiplying the two series and collecting powers:

$$\begin{aligned} x\,\mathrm{g}(x) &= \left(1 - \tfrac{1}{6}x^2 + \tfrac{7}{360}x^4\right)\left(1 - 2x^2 + \tfrac{10}{3}x^4\right) + \cdots \\ &= 1 - \left(2 + \tfrac{1}{6}\right)x^2 + \left(\tfrac{10}{3} + \tfrac{2}{6} + \tfrac{7}{360}\right)x^4 + \cdots \\ &= 1 - \frac{13}{6}x^2 + \frac{1327}{360}x^4 + \cdots, \end{aligned}$$

since $\frac{10}{3} + \frac{1}{3} = \frac{11}{3} = \frac{1320}{360}$.

Checks

- The missing x^3 term in (i) is no accident: $\mathrm{f}(x) - \frac{x}{2} = \ln(\mathrm{e}^{-x/2} + \mathrm{e}^{x/2}) = \ln(2\cosh\frac{x}{2})$ is even, so after $\frac{x}{2}$ only even powers appear.
- Numerically, at $x = 0.1$: $\ln(1+\mathrm{e}^x) \approx 0.7443966600$ and the quartic $\ln 2 + \frac{x}{2} + \frac{x^2}{8} - \frac{x^4}{192}$ agree to about 3.5×10^{-10}.
- At $x = 0.05$, $x\,\mathrm{g}(x)$ and $1 - \frac{13}{6}x^2 + \frac{1327}{360}x^4$ differ by about 9×10^{-8}, roughly $6x^6$, as expected when the next term is in x^6.

Common slips

- Expanding $1/(1+u)$ as $1-u$ only: the u^2 term contributes to x^4 (the $\frac{1}{36}$ and the $4x^4$ above).
- Writing $\cosh 2x = 1 + 2x^2 + \frac{x^4}{24}\cdots$, forgetting that the 4th power of $2x$ is $16x^4$.

Beyond STEP

Part (ii) is about functions with a *pole*: g blows up like $1/x$, and multiplying by x removes the singularity. In complex analysis such a function has a *Laurent series* $\frac{1}{x}\left(1 - \frac{13}{6}x^2 + \cdots\right)$, and the coefficient of $1/x$ (the *residue*, here 1) is the key to evaluating many real integrals. The radius of convergence of the series is set by the nearest other

zero of the denominator, where $\cosh 2x = 0$ at $x = \pm\frac{\pi}{4}\mathrm{i}$. Needham's *Visual Complex Analysis* develops these ideas beautifully.

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